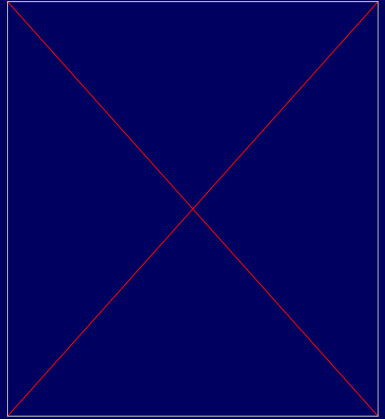
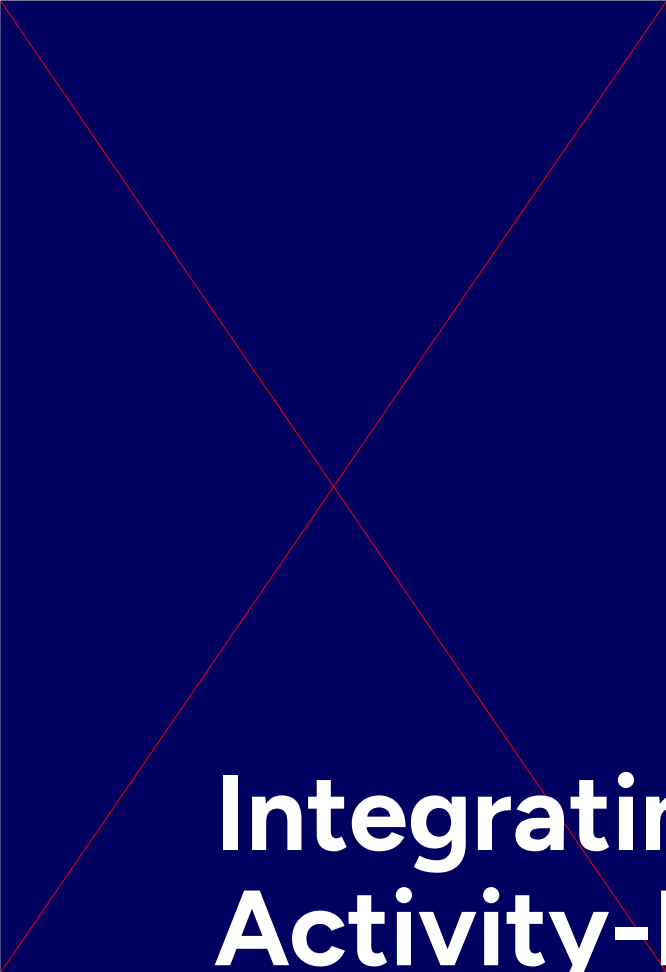
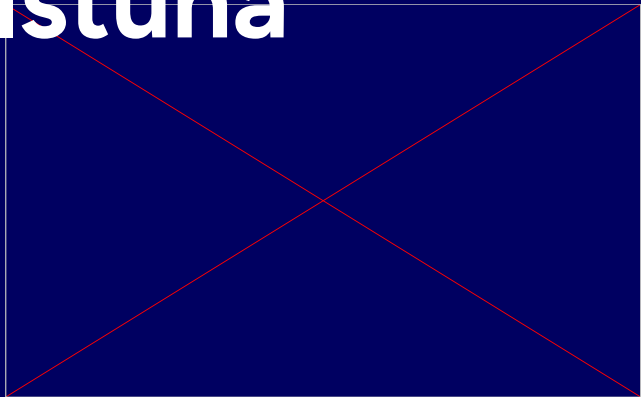
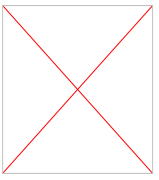




Integrating Bikeability into Network-Based Activity-Based Modeling for Cycling Policy Evaluation: A Case Study of Eskilstuna

Peiling Wu, Fariya Sharmeen, Daniel Jonsson





Introduction and background

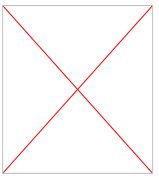
Cities are increasingly promoting cycling to reduce car dependence, emissions, congestion, and public-health burdens.

Why cycling infrastructure needs better evaluation

Cyclists are directly exposed to the conditions of individual street segments and may choose a longer route to obtain greater safety, comfort, or separation from traffic.

Bikeability [1] describes the extent to which the built and natural environment supports safe, comfortable, convenient, and attractive bicycle travel.





Introduction and background

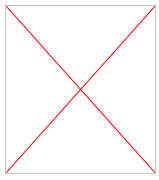
Cities are increasingly promoting cycling to reduce car dependence, emissions, congestion, and public-health burdens.

Why cycling infrastructure needs better evaluation

Where to improve infrastructure

- Most likely to change travel behavior
- Most cost-efficient





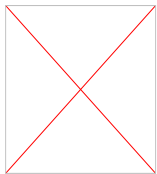
Aim and research questions

To understand the effects of infrastructural changes (bikeability improvements) in facilitating modal shift.

- Using open source tools, e.g. NetAScore and MATSim.
- Developing decision support tools mid sized towns, e.g. Eskilstuna.
- For evidence based targeted sustainable transport policy planning.

Research questions

- How do different infrastructure improvement strategies affect simulated bicycle mode share?
- Which strategy produces the greatest mode shift toward cycling under comparable investment assumptions?
- How to integrate bicycle network and infrastructural measures directly into MATSim to simulate cycling behaviour realistically?



Overall framework

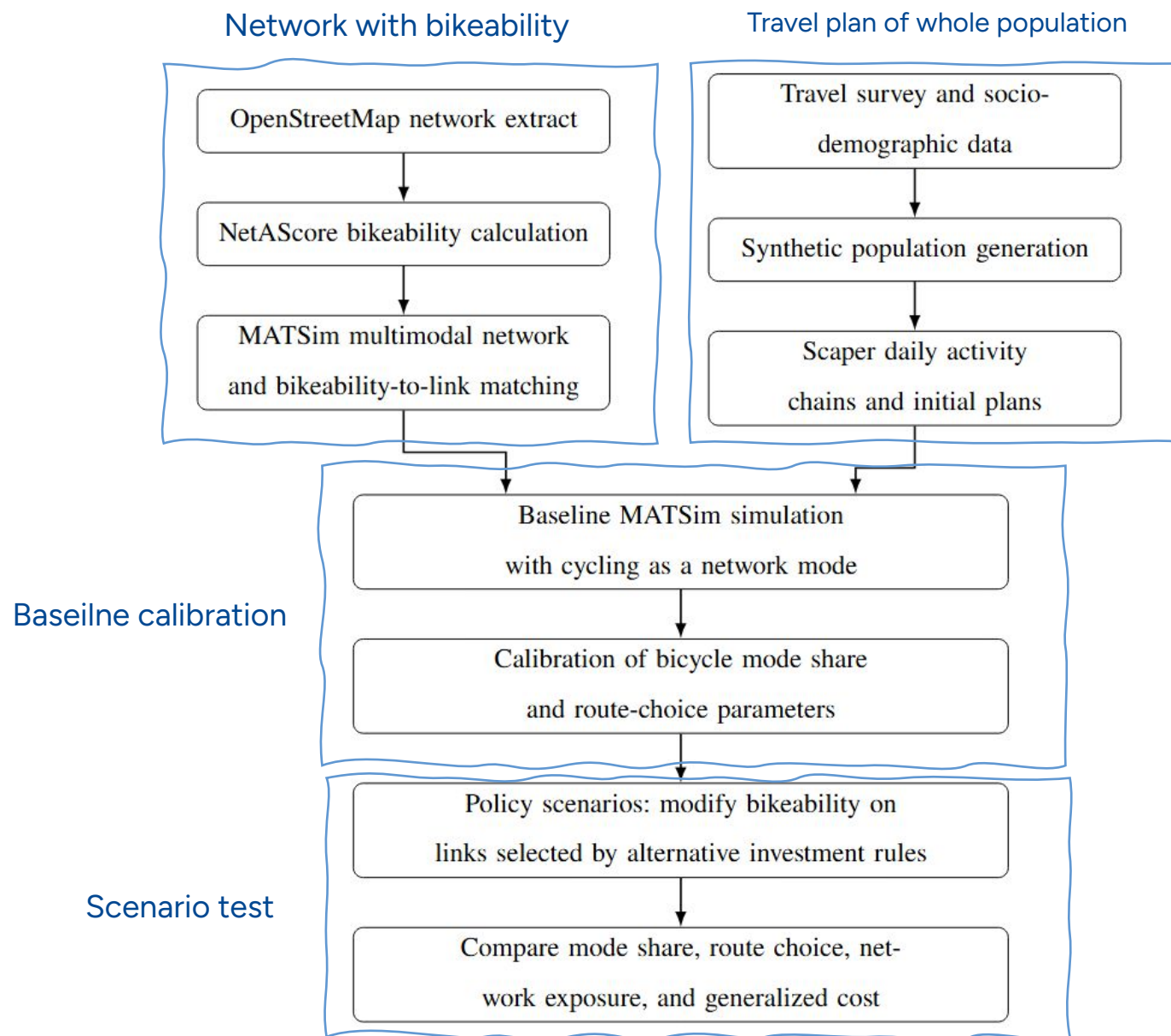
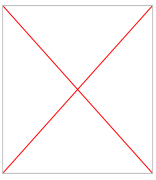


Figure 1 policy evaluation tool framework



Network preparation – bikeability calculation

NetAScore [2] is a toolbox for assessing bikeability and walkability using network datasets.

- Opensource
- Network from OpenStreetMap
- Visualization in QGIS (gpkg)
- Allowing customization and modifying parameters

NetAScore

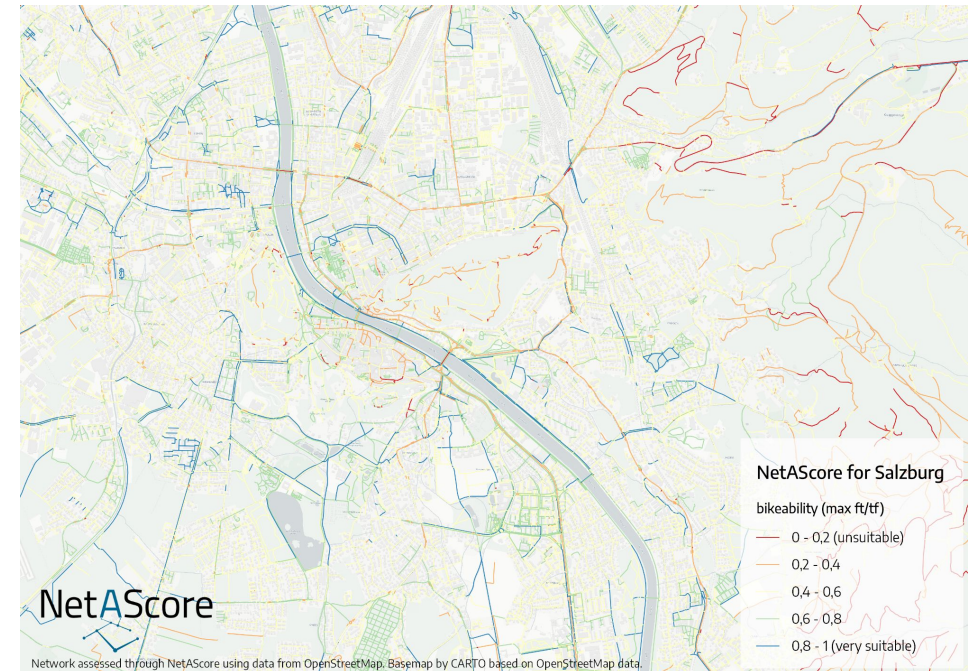
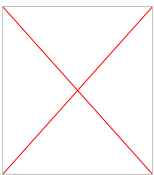
The logo for NetAScore, featuring the word "NetAScore" in a sans-serif font. The "A" is stylized with a blue line graph or network structure integrated into it.

Figure 2 Netascore bikeability example map



Network preparation – bikeability calculation

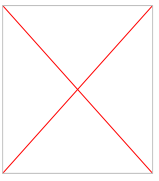
Customized calculation:

1. Customized attributes
2. Customized corresponding weight
3. weighted sum.

$$\text{index for each link} = \frac{\text{weighted sum}}{\text{total weight}}$$

Table 1 Eskilstuna bikeability indicators [3]

Indicator	Weight	Main scoring rule or mapping
Bicycle infrastructure	0.127	Bicycle way = 1; mixed way = 0.9; bicycle/bus lane = 0.75; shared lane = 0.5; undefined = 0.2; no infrastructure = 0.
Road category	0.149	Primary = 0; secondary = 0.2; residential = 0.8; service = 0.85; traffic-calmed = 0.9; no motorized traffic = 1; path = 0.
Maximum speed	0.150	Scores increase as speed decreases: ≥ 100 km/h = 0; ≥ 80 = 0.2; ≥ 70 = 0.3; ≥ 60 = 0.4; ≥ 50 = 0.6; ≥ 30 = 0.85; > 0 = 0.9; 0 = 1.
Parking	0.182	Parking present = 0; no parking = 1.
Crossings	0.128	Crossing-related attributes were included using the NetAScore default bicycle-profile mapping.
Greenness	0.142	Higher greenness receives higher scores: > 75 = 1; > 50 = 0.9; > 25 = 0.8; > 0 = 0.7; 0 = 0.
Other indicators	NULL	Pedestrian infrastructure, designated route, maximum speed greatest, pavement, width, number of lanes, facilities, buildings, water, and noise were retained in the profile but assigned no weight.



Simulate bike trips in MATSim – bicycle extension^[4]

Bicycle network generator

- Link-Based Dynamics: Speed/ route choice could be affected by link attributes
- DEM: impact from slopes
- Customizable Scoring: Enables specific penalties for cycling

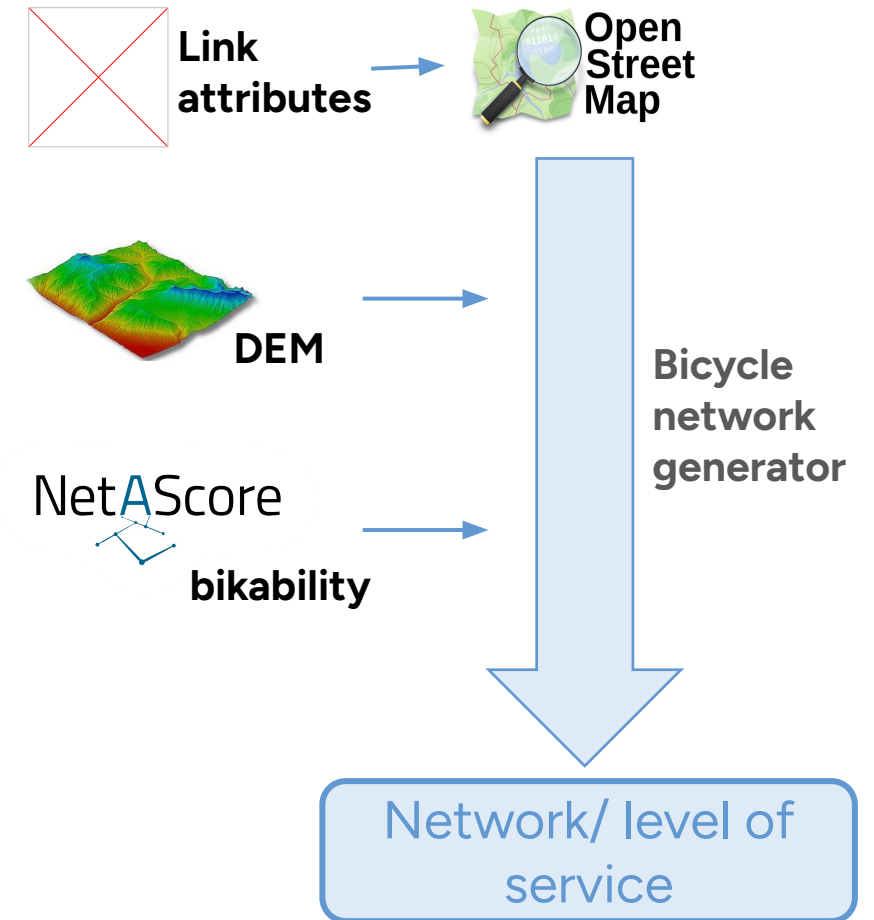
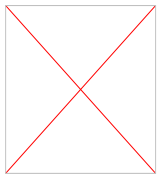


Figure 3 bicycle network construction



Network and travel plan

- Total travel plan: from 55,775 agents, 132,272 trips
- Selected: 30% population, 16,732 agents, 39,967 trips
- 118 zones with population among 120 traffic zones

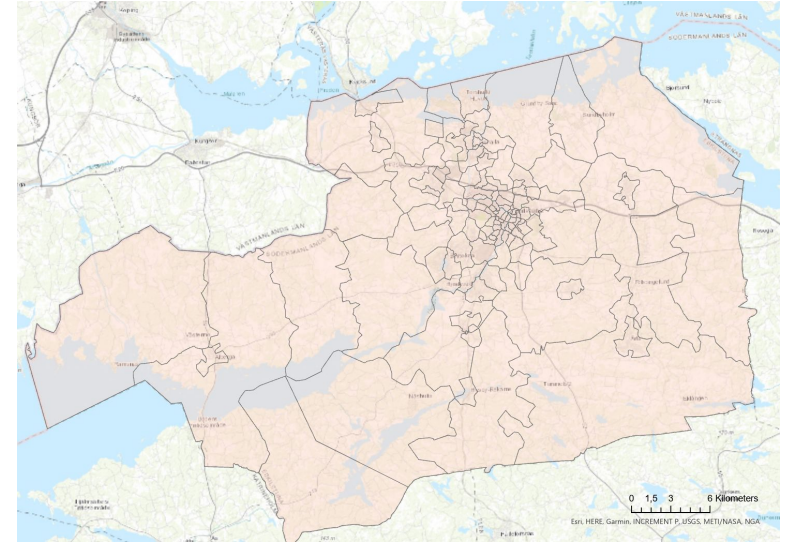


Figure 3 Eskilstuna traffic zone map

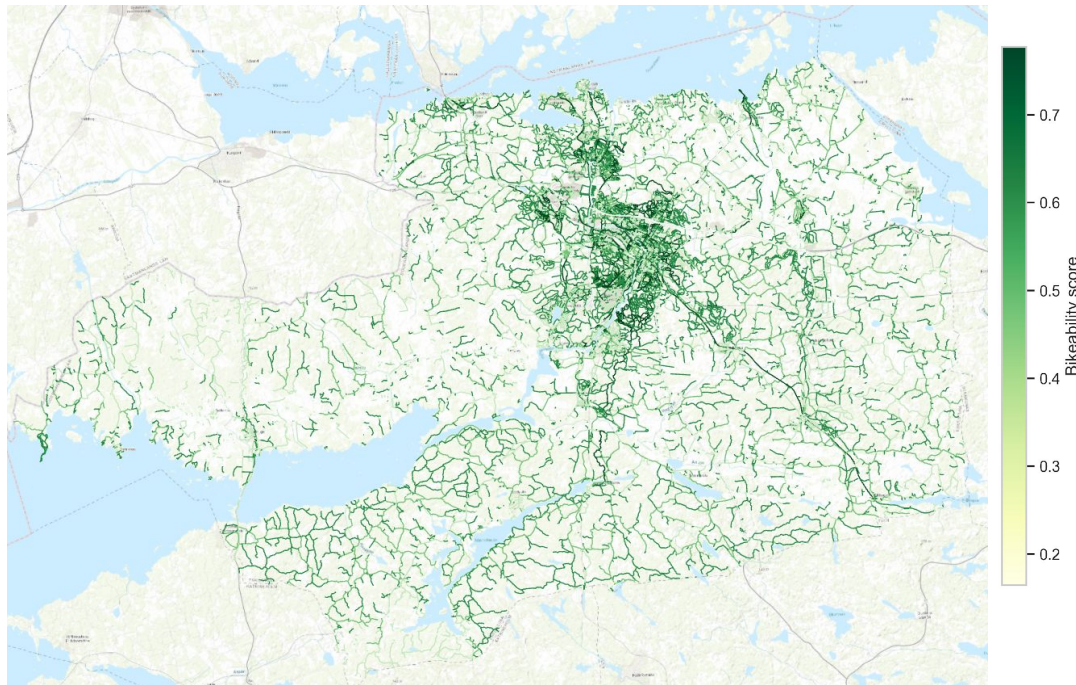


Figure 4 Eskilstuna bikeability spatial distribution

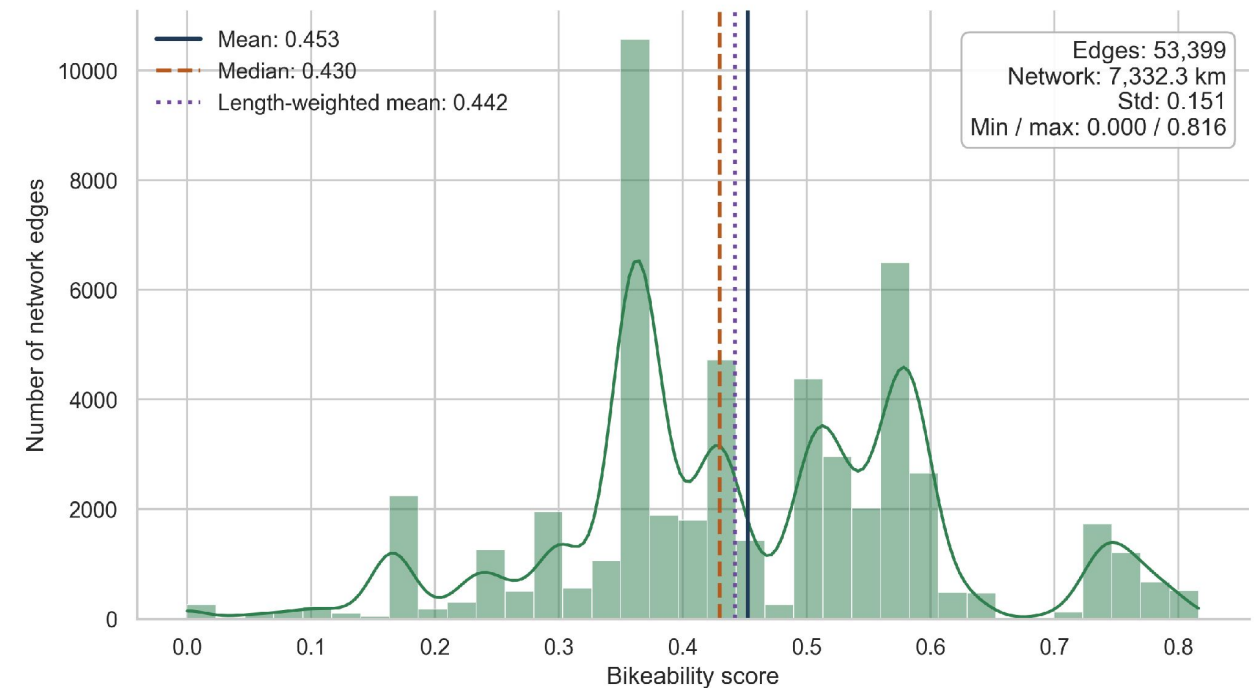
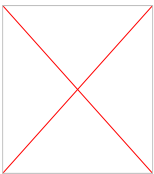


Figure 5 Eskilstuna bikeability value distribution



Network attribute marginal utility	-0.005 util/m
Infrastructure marginal utility	0 util/m
Comfort marginal utility	0 util/m
Gradient coefficient	-0.02 (marginalUtilityOfGradient_m_100m)

Baseline scenario calibration

Calibration: mode share with Eskilstuna travell survey report

Most of the travle utility for bike is included in accessibility marginal utility

Table 2 mode parameters

mode	constant (util)	travel time utility (util/h)	distance utility (util/m)	monetary rate (util/m)
car	-2	-6	-2.5×10^{-4}	-1.5×10^{-4}
bicycle	1.5	-6	-1×10^{-4}	0
walk	0	-6	-5×10^{-4}	0
transit	-1	-6	-4×10^{-4}	-0.5×10^{-4}

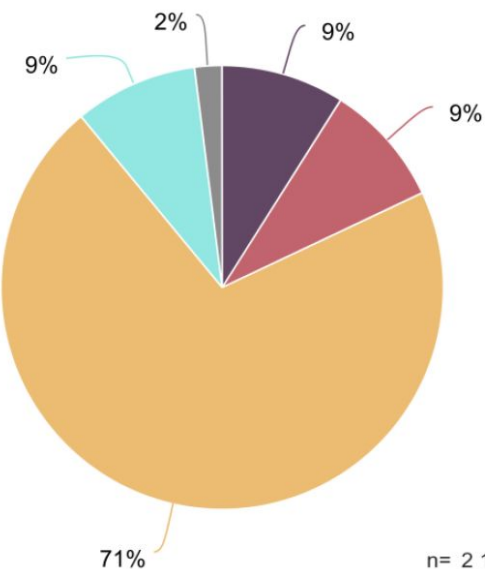


Figure 6 Eskilstuna travel survey mode share

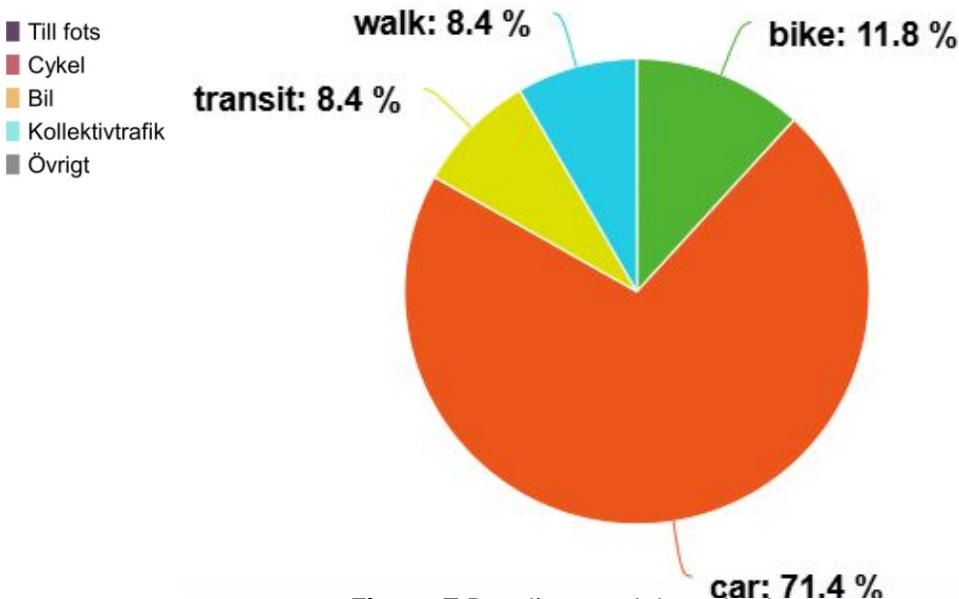
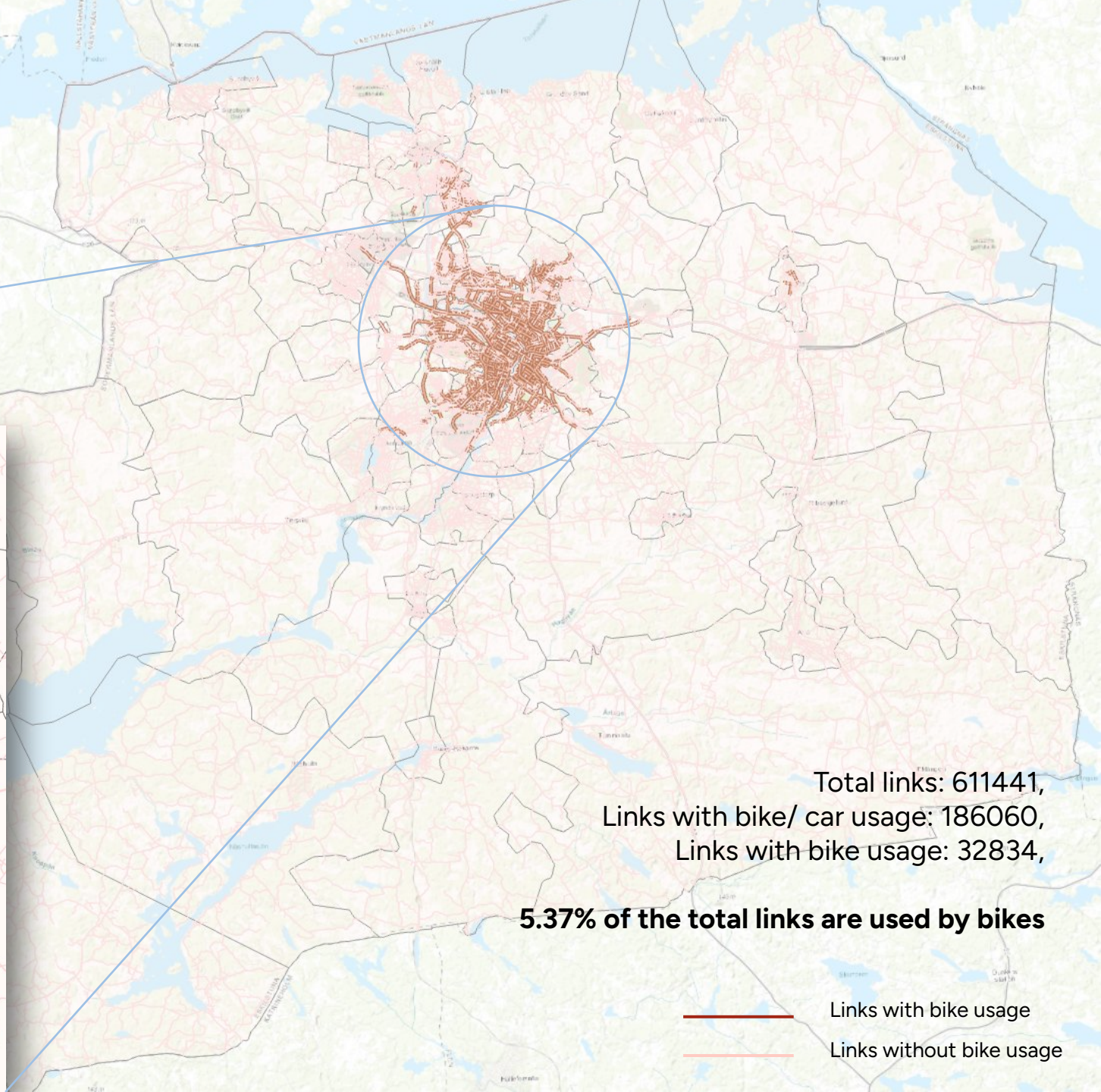
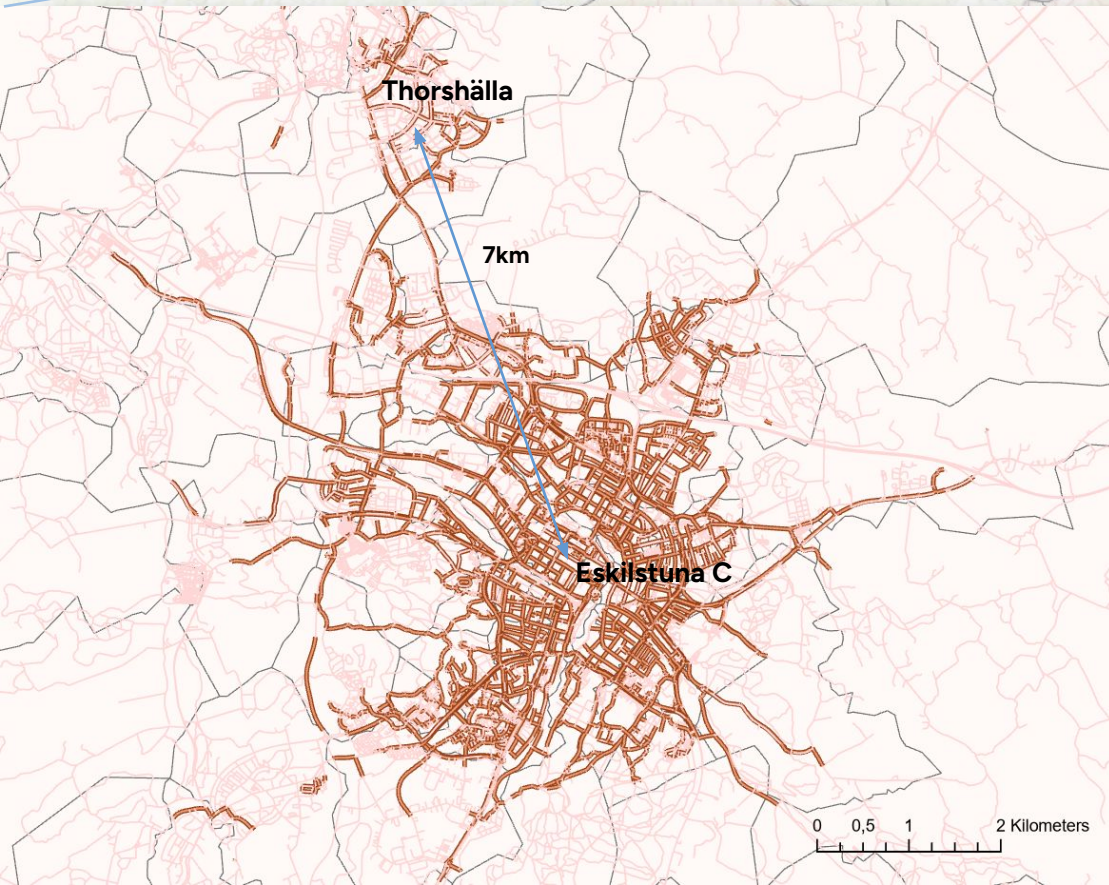
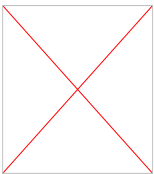


Figure 7 Baseline model mode share

Bike trip distribution





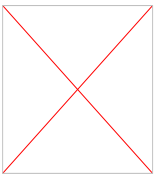
Scenario deisgn

- Four quartiles: ranging from poor to excellent bikeability with the same length

Table 2 bikeability quartiles in Eskilstuna

Quartile	Number of Links	Length (m)	Bicycle Count	Minimum	Median	Maximum
1	6,615	156,355.84	35,154	0.0859	0.3545	0.4434
2	7,956	156,351.90	37,195	0.4434	0.5046	0.5800
3	9,661	156,355.67	183,986	0.5800	0.7366	0.7570
4	8,346	156,380.67	105,709	0.7570	0.7774	0.7978

- S1: upgrading the lowest-bikeability links
- S2: strengthening already high-quality
- S3: improving a randomly selected set of links as a comparison case
- S4: Prioritizing the most frequently used links. In each scenario, improvements are represented by modifying the



Mode share change

- S1: improve poor
- S2: enhance good
- S3: random
- S4: most used

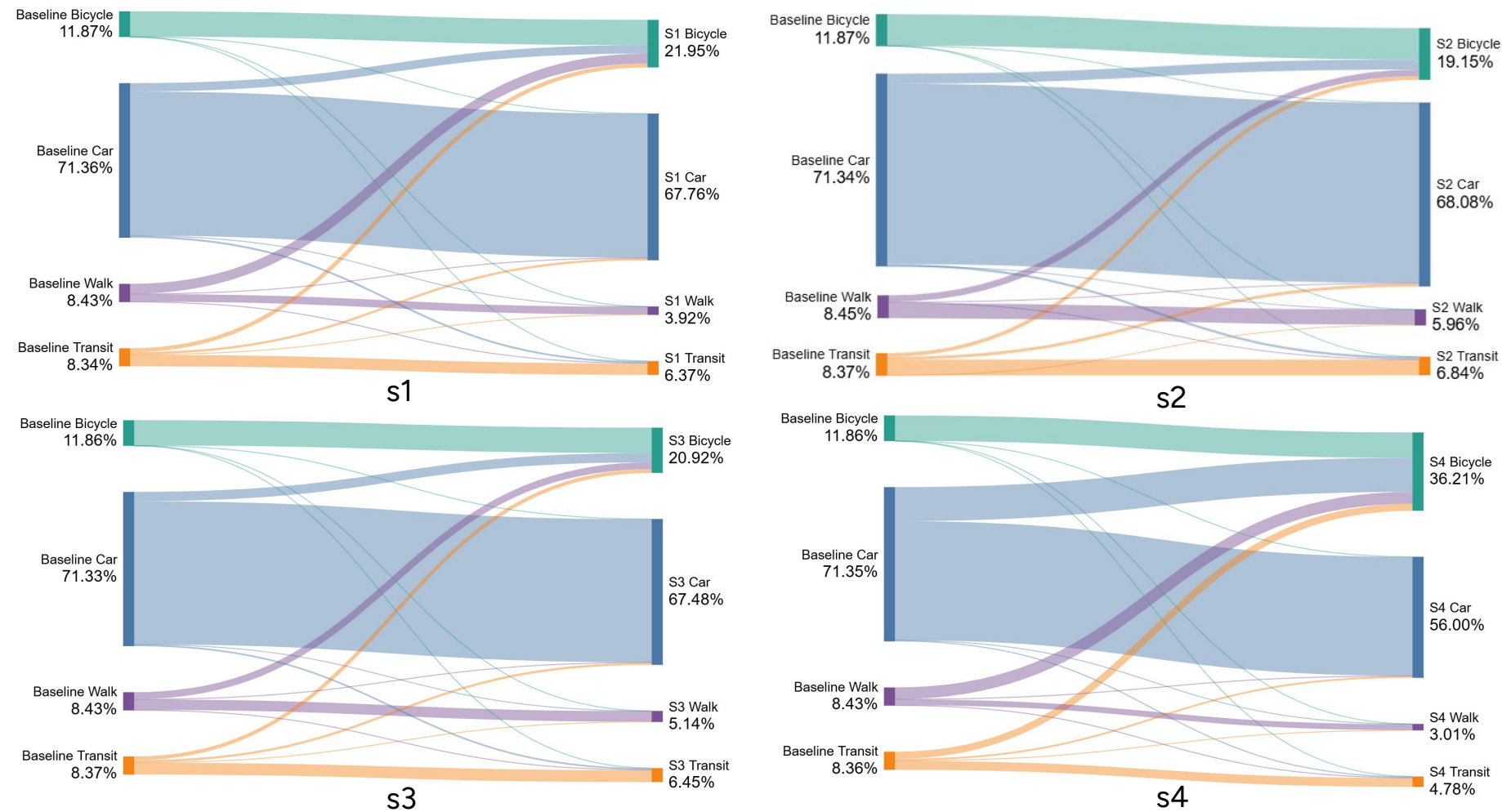
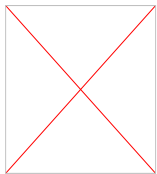


Figure 8 mode share change by scenario



Mode-switched trip distribution

Figure 9 switched car trip distribution

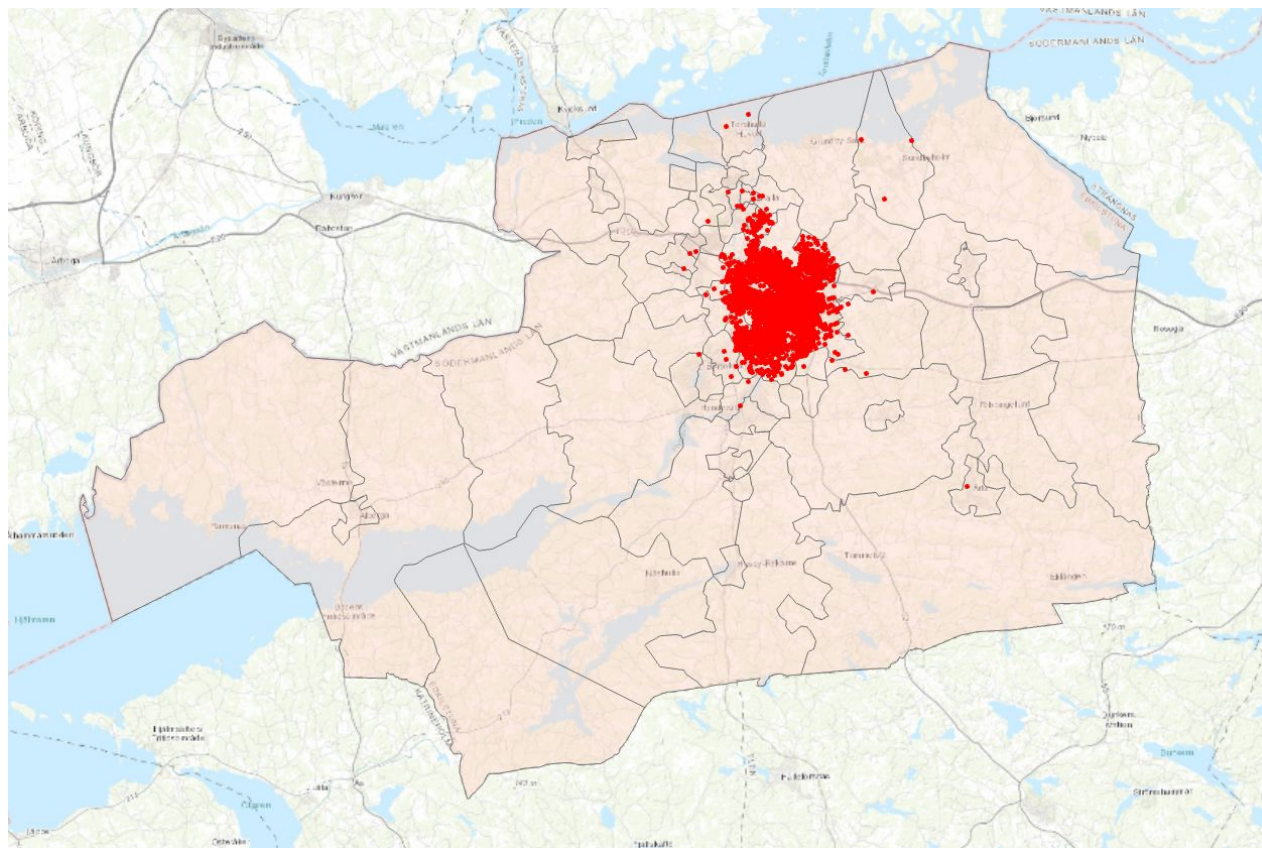
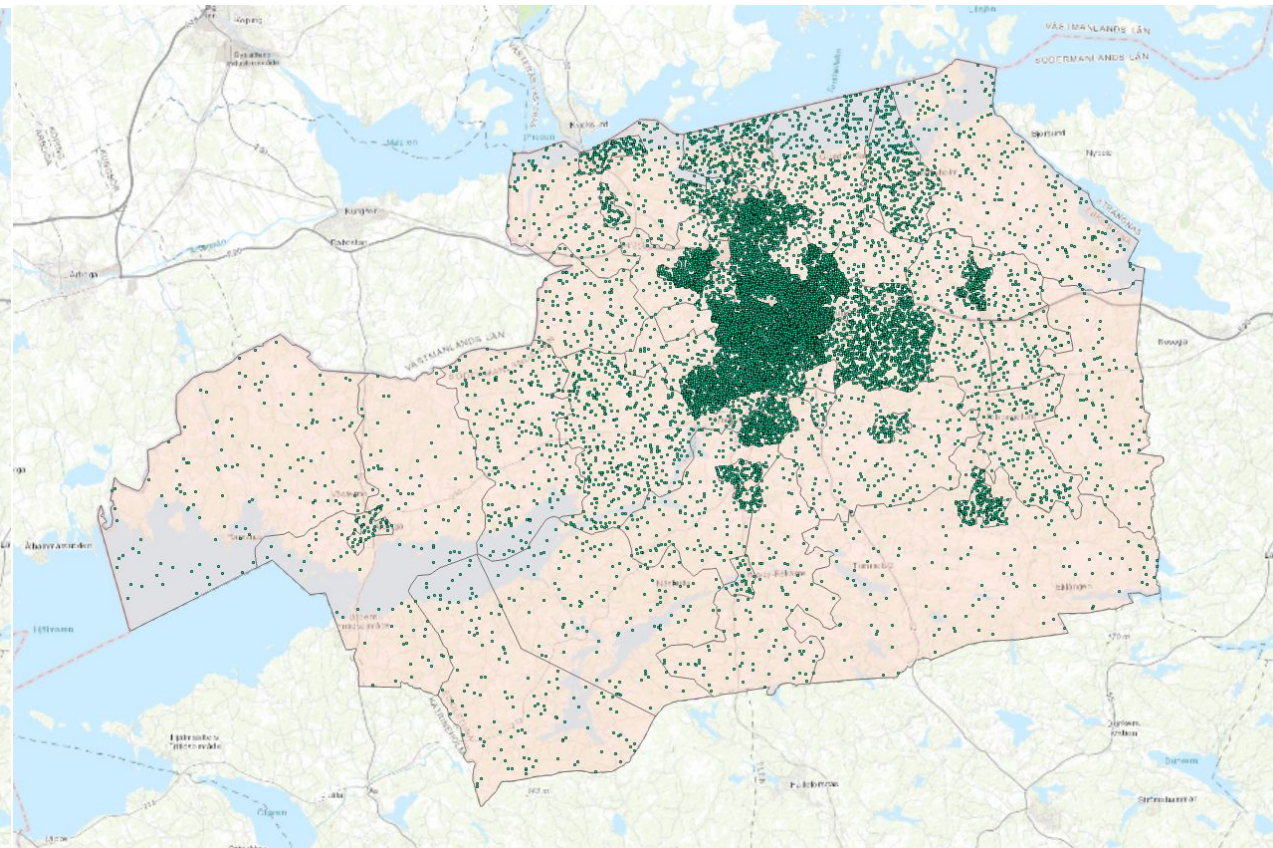
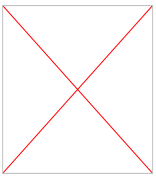


Figure 10 remained car trip distribution





Travel distance change

- The best cycling policy (focusing on the most used ones/ main roads) is not just the one increasing bike trips, but also the one replacing the intended modes, especially short car trips.

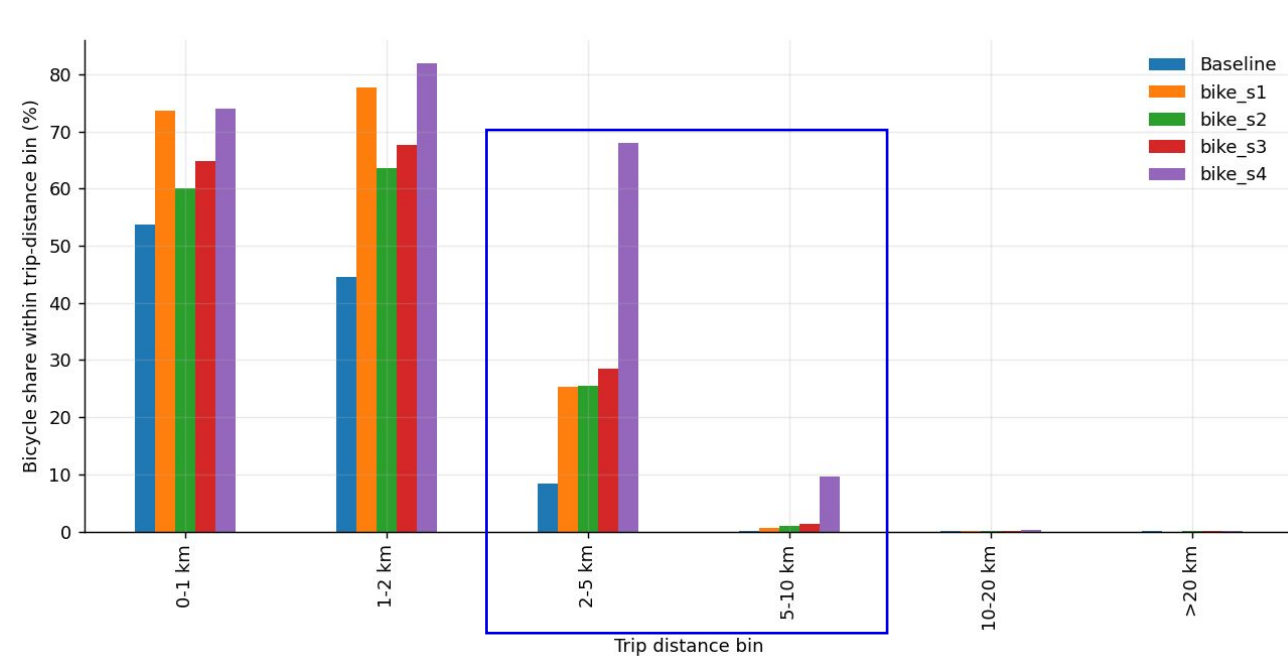


Figure 11 biking distance distribution by scenario

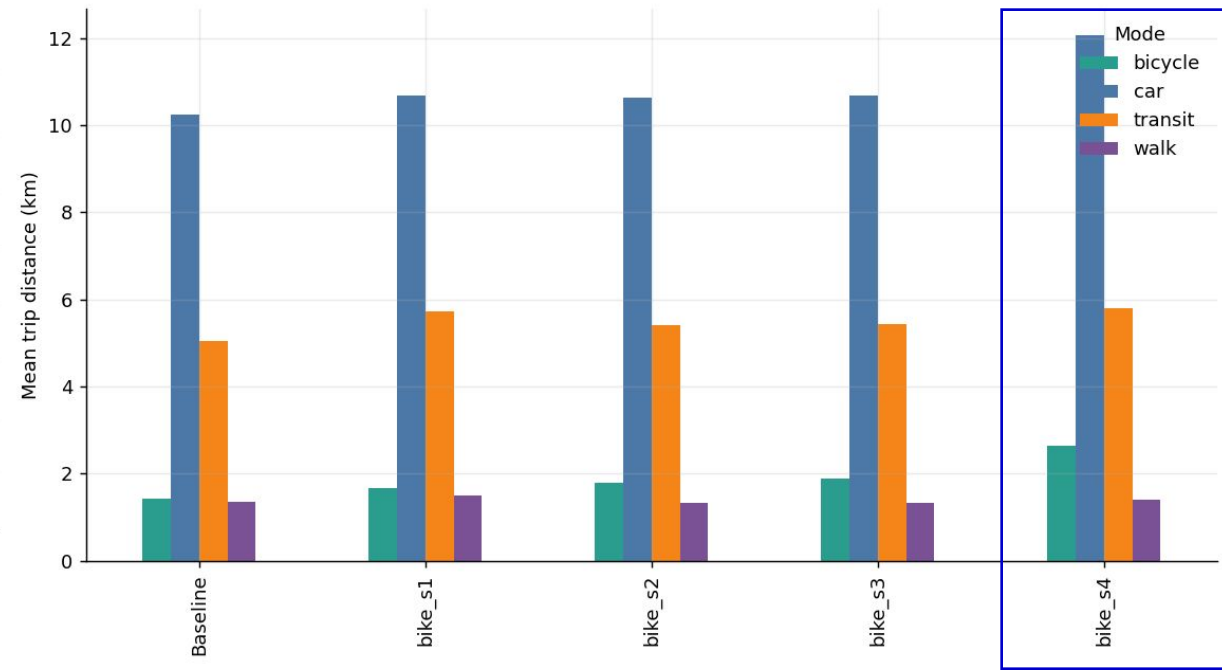
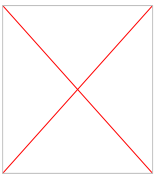


Figure 12 average travel distance by mode

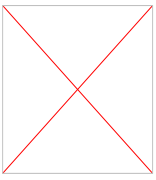


Travel distance change

- Car to bike trip: trips with mode switch is shorter

Table 3 car-bike shift trips and average travel distance/ time

scenario	group	Baseline distance km	Scenario distance km	Distance change km	Baseline time min	Scenario time min	Time change min
S1	changer	2.62	2.37	-0.25	6.02	11.63	5.61
S1	stayer	10.73	10.73	\	21.21	21.37	\
S2	changer	2.88	2.65	-0.23	6.45	13.27	6.82
S2	stayer	10.69	10.69	\	21.15	21.17	\
S3	changer	3.01	2.76	-0.25	6.75	13.78	7.03
S3	stayer	10.75	10.75	\	21.26	21.26	\
S4	changer	3.74	3.57	-0.17	8.14	17.68	9.55
S4	stayer	12.12	12.11	\	23.84	23.89	\



Travel distance & time change

Compare trips switched from car to bike:

- S1-S3: Most trips are less than 5 km
- S4: most trips are less than 7km
- S1-S3: Most time differences are less than 15 min
- S4: Most time differences are less than 20 min

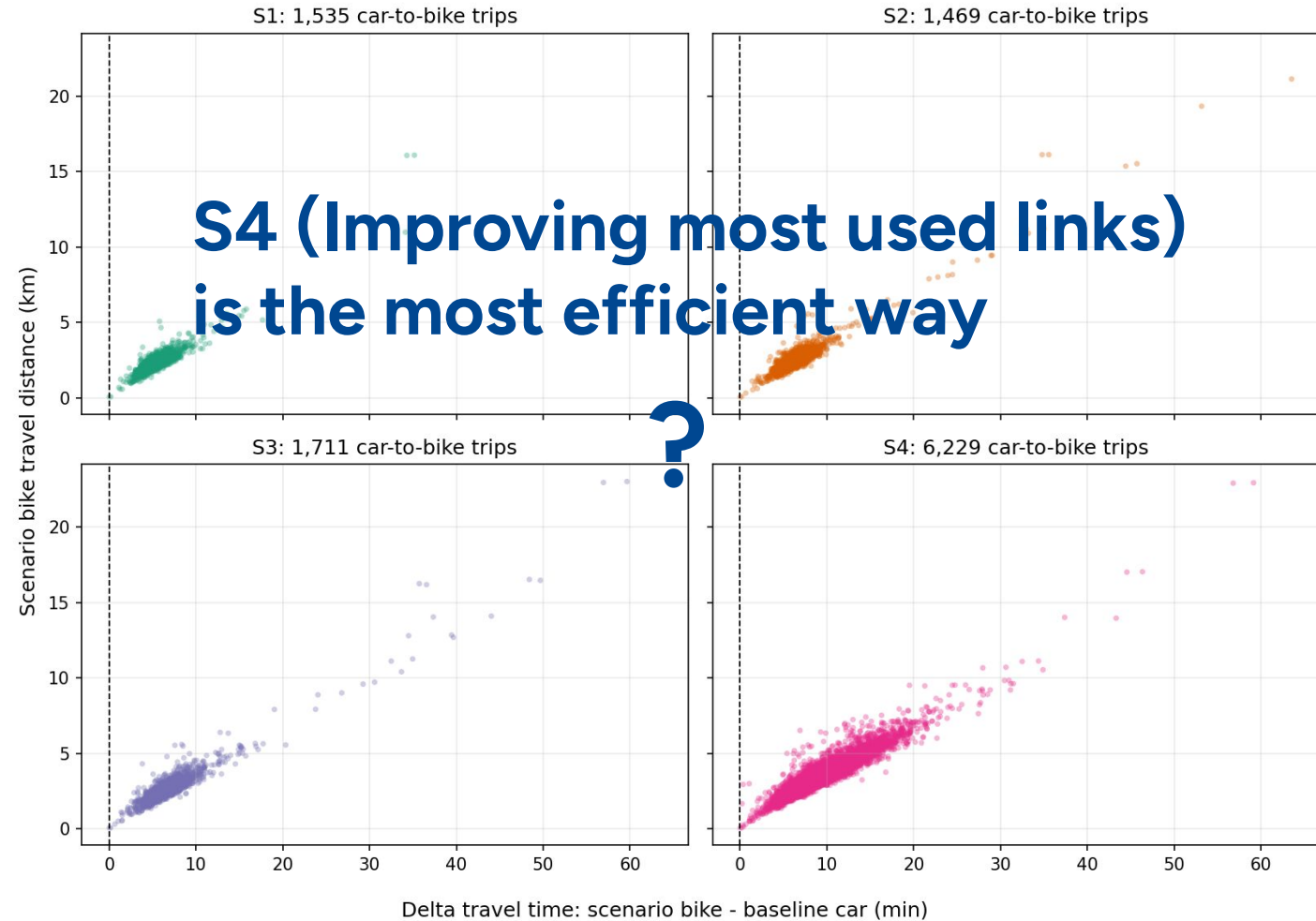
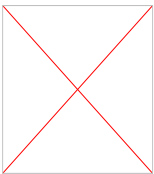


Figure 13 travel distance and time change of switched trips



Effect from changing route?

Hypothesis: not enough agents test the improved links in S1-3, resulting in lower mode change rate

Replanning strategy:

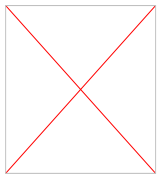
- 300 iterations learning rate 0.8
 - Plan Selector 0.45
 - Plan Mutator (mode) 0.45
 - Plan Mutator (route) 0.1
- 0.45

Table 4 MATSim setup

Plan scoring and replanning		
Learning and selection	Learning rate; BrainExpBeta	0.8; 2.0
Activity scoring	Performing; late arrival	6; -18 util/h
Other time scoring	Early departure; waiting; PT waiting	0; 0; -2 util/h
Money	Marginal utility of money	1 util/monetary unit
Strategy weights	ChangeExpBeta / SubtourModeChoice / ReRoute	0.45 / 0.45 / 0.10
Plan memory	Maximum stored plans per agent	5
Innovation cutoff	Fraction to disable innovation	0.8
Mode-choice alternatives	Available modes	Car, bicycle, walk, transit
Vehicle-chain constraints	Chain-based modes	Car, bicycle
Car availability	Consider car availability	False
Single-trip mode changes	Random single-trip mode probability	0.1

Table 5 comparison – rerouting stragety

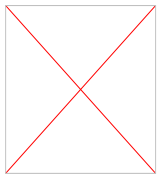
	bicycle	car	transit	walk
0.1 s1	21.92%	67.78%	6.38%	3.93%
0.45 s1	22.13%	68.25%	5.87%	3.76%
0.1 s2	19.11%	68.10%	6.82%	5.96%
0.45 s2	19.14%	68.78%	6.27%	5.81%



Conclusion

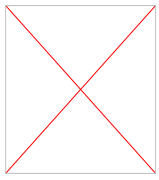
The framework helps identify not only where bikeability is low, but which improving strategy is most efficient in changing travel behavior.

- Promoting biking infrastructure contributes to increasing long distance bike trips
- Bike trips and increased bike trips are concentrated around Eskilstuna city center
- Compared to strategies based on original infrastructure, improving most used links is more efficient for improving biking rate
- **Contributions**
 - Links NetAScore bikeability assessment with MATSim bicycle simulation.
 - Evaluates alternative cycling infrastructure scenarios using a common activity-based demand model.
 - Mode-shift-focused indicators support more targeted cycling infrastructure planning.



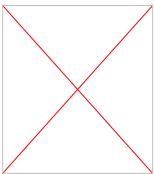
Limitation

- Clearer breakdown of bikeability weights & implementation.
- Calibration data limitations (Need for observed bicycle counts/GPS validation)
- Data in city outskirts (travel survey and road parameter)



References

- [1]. Kellstedt, D. K., Spengler, J. O., Foster, M., Lee, C., & Maddock, J. E. (2021). A scoping review of bikeability assessment methods. *Journal of community health*, 46(1), 211-224.
- [2]. Werner, C., Wendel, R., Kaziyeva, D., Stutz, P., Van der Meer, L., Lackner, B. M., ... & Loidl, M. (2022). NetAScore: Network Assessment Score Toolbox for Sustainable Mobility.
- [3]. Ivana Paulusová, Fariya Sharmeen. (2025). School bikeability index: A case study of primary schools in Stockholm, *Journal of Cycling and Micromobility Research*, Volume 5.
- [4]. Ziemke, D., Metzler, S., & Nagel, K. (2019). Bicycle traffic and its interaction with motorized traffic in an agent-based transport simulation framework. *Future Generation Computer Systems*, 97, 30-40.



Thank you!